

Modeling Disease Networks using SIR Simulation

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Understanding how infectious diseases spread through human populations remains a central challenge in public health, particularly in highly connected social networks. This project investigates disease propagation using a network-based modeling framework, with the primary research question: which group of individuals is most susceptible to infection, what is the minimum fraction of this group that must be immunized to significantly reduce or suppress disease spread, and what strategies can we use to suppress infection and sickness spread the fastest? Identifying targeted vaccination strategies is especially important because widespread immunization can be costly, slow, or logistically difficult, while poorly targeted interventions may fail to prevent outbreaks. The study models a social network derived from online or real-world interaction data, where individuals are represented as nodes and social interactions as weighted edges. Edge weights reflect contact frequency or transmission likelihood. Additional attributes such as disease contagiousness, recovery or mortality rates, and individual vulnerability are incorporated into the model. Using this structure, we simulate disease transmission using random-walk-based spreading and diffusion over the graph. Spectral properties of the network, particularly those derived from the graph Laplacian, are analyzed to understand connectivity, cluster structure, and the potential speed and reach of outbreaks. Through repeated simulation and analysis, we expect to identify that highly connected individuals (high-degree nodes), along with those possessing higher intrinsic vulnerability to infection, play a disproportionately important role in causing virus transmission. Targeted immunization of a relatively small subset of these individuals may dramatically reduce overall spread.

Disease Networks | SIR Simulation | Epidemic Modeling | Targeted Immunization

Introduction

The spread of infectious diseases remains a major public health concern, especially in highly connected social environments. Modern patterns of interaction create complex networks through which infections can propagate rapidly. Traditional epidemiological models often assume uniform mixing, which fails to capture the heterogeneous structure of real-world social networks. Network-based approaches provide a more realistic framework by representing individuals as nodes and interactions as edges, allowing researchers to analyze how connectivity and clustering influence disease transmission. Vaccination is one of the most effective methods for preventing outbreaks, but large-scale immunization campaigns can be costly and difficult to implement. In many situations, limited resources require targeted intervention strategies. Understanding which individuals or groups contribute most to disease spread is therefore essential for designing efficient control policies. This project investigates which groups of individuals in a social network are most susceptible to infection and what minimum fraction of these groups must be immunized to suppress disease spread. Using weighted network models, random-walk-based simulations, and spectral analysis of the graph Laplacian, we analyze how structural properties affect transmission dynamics. We hypothesize that highly connected and highly vulnerable individuals play a dominant role in spreading

Significance

This study provides insights into how the structure of social networks influences the spread of infectious diseases and highlights the critical need for targeted immunization strategies to efficiently suppress outbreaks. By utilizing a network-based framework and spectral analysis, the research identifies the disproportionate impact of highly connected and vulnerable individuals in transmission dynamics.

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disease, and that targeted immunization of a relatively small subset of these nodes can significantly reduce overall transmission.

Expected Results

We expect the simulations to demonstrate that disease spread is not homogeneous across a real world graph, with infection risk among specific subgroups of individuals rather than being uniformly distributed. In particular, nodes with high degree and high weighted connectivity are anticipated to exhibit higher infection rates. Spectral analysis of the graph Laplacian is expected to reveal that tightly connected clusters facilitate rapid local spread, while bridge nodes connecting clusters control for global diffusion. Specifically networks with smaller spectral gaps are expected to show faster outbreaks and broader reach, whereas networks with pronounced community structure may exhibit localized outbreaks that take longer to propagate. Simulation results are anticipated to indicate that random immunization strategies require a relatively large fraction of the population to be vaccinated to achieve meaningful suppression of disease spread. In contrast, targeted immunization that lowers the spectral gap is expected to produce a disproportionate reduction in outbreak size and transmission speed.

Mathematical Problem Statement

Let $G = (V, E)$ be a contact network with $|V| = n$ individuals. We run a discrete-time stochastic SIR process on G with transmission rate β and recovery rate γ .

Each node $v \in V$ has state $X_v(t) \in \{S, I, R\}$ at time t . For each time step:

- If $X_v(t) = S$, then v becomes infected with probability

$$p_v(t) = 1 - \exp\left(-\beta \sum_{u \in N(v)} w_{uv} \mathbf{1}\{X_u(t) = I\}\right),$$

where w_{uv} is the contact weight on edge (u, v) .

- If $X_v(t) = I$, then v recovers with probability γ .
- Recovered nodes remain in state R.

Before the epidemic starts, choose an immunized set $U \subseteq V$ with budget

$$|U| = k, \quad 0 \leq k \leq n,$$

and initialize all $v \in U$ in state R.

For a fixed immunization set U , let $I_t(U)$ be the number of infected nodes at time t , and define

$$I_{\max}(U) = \max_t I_t(U),$$

$$T_{\inf}(U) = |\{v \in V : v \text{ becomes infected at least once}\}|.$$

The optimization goal is to choose U to minimize either expected peak prevalence

$$\min_{U \subseteq V, |U|=k} \mathbb{E}[I_{\max}(U)],$$

or expected outbreak size

$$\min_{U \subseteq V, |U|=k} \mathbb{E}[T_{\inf}(U)].$$

In practice, the expectations are estimated with Monte Carlo trials, and strategies (random, degree-based, centrality-based, nomination-based, adaptive) are compared as candidate constructions of U .

Limitations

Key risks include the high computational cost of eigenvalue decomposition on extremely large networks and the potential for "noise" or missing edges in real-world datasets, which might obscure the transmission dynamics.

Materials and Methods

This project uses publicly available network data source frameworks to ensure reproducibility and methodological rigor in the following subsections:

- **High-Resolution Synthetic Contact Networks**
- **Empirical Social Interaction Structure**
- **Established Propagation Models**
- **Centrality-Based Intervention Strategies**

1. High-Resolution Synthetic Contact Networks. We use datasets representing individual-level interactions within households and neighborhoods. These networks simulate realistic, multi-layered, heterogeneous mixing patterns across hundreds of thousands of nodes. These datasets are commonly used in epidemiological modeling of disease spread in these environments.

2. Empirical Social Interaction Structure. The contact network is derived from shared environments (e.g., occupation, location, etc.), allowing us to model transmission along socially meaningful pathways beyond homogeneous mixing.

3. Established Propagation Models. Disease spread is modeled using the SIR (Susceptible-Infected-Recovered) framework. The model dynamics, where infected nodes may infect multiple neighbors, capture effects when multiple infected neighbors attempt to transmit.

4. Centrality-Based Intervention Strategies. To evaluate the impact of graph-theoretic centrality measures to identify structurally important nodes for intervention.

- **Degree centrality:** identifies locally high-contact "hubs".
- **Eigenvector centrality:** identifies globally influential nodes.

These measures allow for the comparison of targeted immunization strategies. All computational methods rely on widely available software packages, such as **networkx** and **numpy**, ensuring the accessibility and reproducibility of the framework.

Data, Materials, and Software Availability. All data and code are available in an open-source repository.

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